DPP No: 06

Maximum Time 50 Min

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C³

(COMPETISHUN CRASH COURSE)

PHYSICS

TARGET JEE-MAINS

SYLLABUS : WORK POWER & ENERGY

1. The work done by the frictional force on a pencil in drawing a complete circle of radius $r = 1/\pi$ metre on the surface by a pencil of negligible mass with a normal pressing force $N = 5$ N ($\mu = 0.5$) is :

(A) + 4J (B) –3 J (C) – 2 J (D) – 5J

2. A rigid body of mass m is moving in a circle of radius r with a constant speed v. The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre. What is the work done by this force in moving the body over half the cirumference of the circle.

(A) $\frac{mv^2}{\pi r^2}$ (B) Zero (C) $\frac{mv^2}{r^2}$ (D) $\frac{\pi r^2}{mv^2}$

3. A man pushes wall and fails to displace it. He does

(A) Negative work (B) Positive but not maximum work

(C) No work at all (D) Maximum work

4. A 50 kg man with 20 kg load on his head climbs up 20 steps of 0.25 m height each. The work done by the man on the block during climbing is

(A) 5 J (B) 350 J (C) 1000 J (D) 3540 J

5. A ball is released from the top of a tower. The ratio of work done by force of gravity in first, second and third second of the motion of the ball is

(A) 1 : 2 : 3 (B) 1 : 4 : 9 (C) 1 : 3 : 5 (D) 1 : 5 : 3

6. A particle moves under the effect of a force $F = Cx$ from $x = 0$ to $x = x_1$. The work done in the process is

(A) Cx_1^2 (B) $\frac{1}{2}Cx_1^2$ (C) Cx_1 (D) Zero

7. A rigid body is acted upon by a horizontal variable force which is inversely proportional to the distance covered from its initial position 's'. The work done by this force will be proportional to :

(A) s (B) s^2 (C) $\sqrt{s}$ (D) None of these

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8. A force acting on a particle varies with the displacement x as $F = ax - bx^2$. Where a = 1 N/m and $b = 1\ N/m^2$. The work done by this force for the first one meter (F is in newtons, x is in meters) is:

(A) $\frac{1}{6}$J (B) $\frac{2}{6}$ J (C) $\frac{3}{6}$J (D) None of these

9. A particle moves along the x-axis from x = 0 to x = 5 m under the influence of a force F(in N) given by $F = 3x^2 - 2x + 7$. Calculate the work done by this force.

(A) 75 J (B) 115 J (C) 125 J (D) 135 J

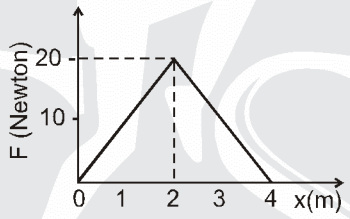
10. A person A of 50 kg rests on a swing of length 1m making an angle 37° with the vertical. Another person B pushes him to swing on other side at 53° with vertical. The work done by person B is : [$g = 10\ m/s^2$]

(A) 50 J (B) 9.8 J (C) 100 J (D) 10 J

11. The kinetic energy of a body of mass 2 kg and momentum of 2 Ns is

(A) 1 J (B) 2J (C) 3 J (D) 4 J

12. The graph between the magnitude of resistive force F acting on a body and the position of the body travelling in a straight line is shown in the figure. The mass of the body is 25 kg and initial velocity is 2 m/s. When the distance covered by the body is 4m, its kinetic energy would be (not other force acts on it)



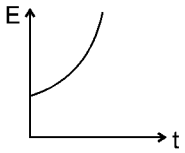
(A) 50 J (B) 40 J (C) 20 J (D) 10 J

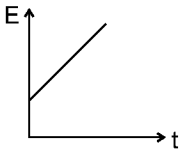
13. Work done in time t on a body of mass m which is accelerated from rest to a speed v in time t_1 as a function of time t is given by :

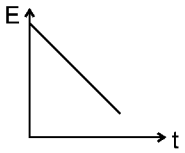
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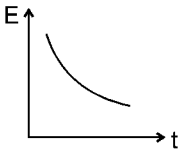
(A) $\frac{1}{2}m\frac{v}{t_1}t^2$ (B) $m\frac{v}{t_1}t^2$ (C) $\frac{1}{2}\left(\frac{mv}{t_1}t\right)^2 t^2$ (D) $\frac{1}{2}m\frac{v^2}{t_1^2}t^2$

14. A particle is projected horizontally from a height h. Taking g to be constant every where, kinetic energy E of the particle with respect to time t is correctly shown in (Neglect air resistance)

(A) 

(B) 

(C) 

(D) 

15 A small sphere starts falling from a very large height and after falling a distance of 100 m it attains the terminal velocity and continues to fall with this velocity. The work done by the atmosphere during the first fall of 100m is :

(A) Greater than the work done for next fall of 100 m

(B) Less than the work done for next fall of 100 m

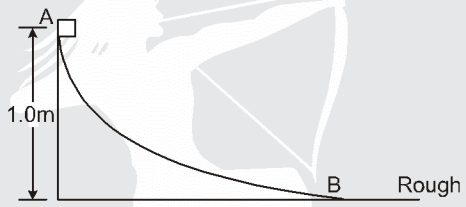
(C) Equal to 100 mg

(D) Greater than 100 mg

16. A retarding force is applied to stop a train. The train stops after 80 m. If the speed is doubled, then the distance travelled when same retarding force is applied is

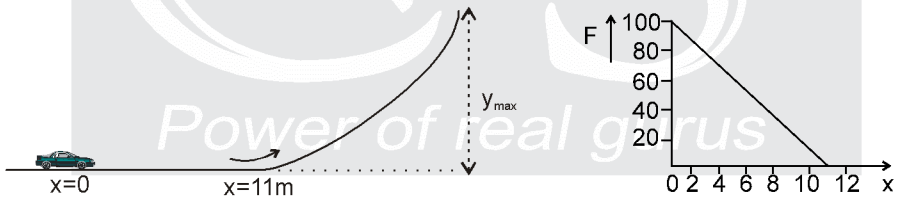
(A) The same (B) Doubled (C) Halved (D) Four times

17. A block weighing 10 N travles down a smooth curved track AB joined to a rough horizontal surface (figure). The rough surface has a friction coefficient of 0.20 with the block. If the block starts slipping on the track from a point 1.0 m above the horizontal surface, the distance it will move on the rough surface is :



(A) 5.0 m (B) 10.0 m (C) 15.0 m (D) 20.0 m

18. A toy car of mass 5 kg starts from rest and moves up a ramp under the influence of force F (F is applied in the direction of velocity) plotted against displacement x. The maximum height attained is given by ($g = 10\ m/s^2$)



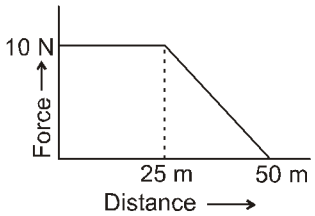
(A) $y_{max} = 20$ m (B) $y_{max} = 15$ m (C) $y_{max} = 11$ m (D) $y_{max} = 5$ m

19. A force of 1000 N acts on a particle parallel to its direction of motion which is horizontal. Its velocity increases from 1 m s^{-1} to 10 m s^{-1}, when the force acts through a distance of 4 metre. Calculate the mass of the particle. Given : a force of 10 Newton is necessary for overcoming friction.

(A) 30 kg (B) 50 kg (C) 80 kg (D) 105 kg

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20. A body of mass 5 kg is acted upon by a variable force. The force varies with the distance covered by the body. What is the speed of the body when the body has covered 25 m? Assume that the body starts from rest.



(A) 5 m/s (B) 10 m/s

(C) 15 m/s (D) 25 m/s

21. A block of mass 200 g is moving with a speed of 4 m/s at the highest point in a closed circular tube of radius 10 cm kept in a vertical plane. The cross-section of the tube is such that the block just fits in it. The block makes several oscillations inside the tube and finally stops at the lowest point. Find the work done by the tube on the block during the process. ($g = 10\ m/s^2$)

(A) – 2 J (B) – 1 J (C) + 2 J (D) 2 J

22. A weight is hung freely from the end of a spring. A boy then slowly pushes the weight upwards until the spring becomes slack. The gain in gravitational poetential energy of the weight during this process is equal to :

(A) The work done by the boy against the gravitational force acting on the weight.

(B) The loss of the stored energy by the spring minus the work done by the tension in the spring.

(C) The work done on the weight by the boy plus the stored energy lost by the spring.

(D) The work done on the weight by the boy minus the workdone by the tension in the spring plus the stored energy lost by the spring.

23. The negative of the work done by the conservative internal forces on a system equals the change in its

(A) total energy (B) kinetic energy (C) potential energy (D) none of these

24. A stone is projected vertically up with a velocity u, reaches upto a maximum height h. When it is at a height of 3h/4 from the ground, the ratio of KE and PE at that point is : (consider PE = 0 at the point of projection)

(A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 3 : 1

25. Two springs A and B ($k_A = 2k_B$) are stretched by applying forces of equal magnitudes at the four ends. If the energy stored in A is E, then in B is (assume equilibrium):

(A) E/2 (B) 2E (C) E (D) E/4

ANSWER KEY

1.	(D)	**2.**	(B)	**3.**	(C)	**4.**	(C)	**5.**	(C)
6.	(B)	**7.**	(D)	**8.**	(A)	**9.**	(D)	**10.**	(C)
11.	(A)	**12.**	(D)	**13.**	(D)	**14.**	(A)	**15**	(B)
16.	(D)	**17.**	(A)	**18.**	(C)	**19.**	(C)	**20.**	(B)
21.	(A)	**22.**	(C)	**23.**	(C)	**24.**	(C)	**25.**	(B)